

PDAN8411

Njabulo Mbulelo Nkosi – ST10299524
PROGRAMMING FOR DATA ANALYTICS 1 FINAL POE

Table of Contents

Introduction	2
Organising the Analysis.....	2
Review of the Dataset.....	2
Exploratory Data analysis.....	3
Distribution of Classes.....	3
Analysing Review Length	3
Word Clouds.....	4
Pre-processing of text.....	5
TF-IDF feature extraction	5
Cross-Validation and Model Selection	5
Hyperparameter Optimisation.....	5
Evaluation of the Model Using the Test Set.....	6
Retraining with Different Parameters.....	7
Predictive System.....	7
Feature importance: The Best Words for Each Class.....	7
In the end.....	8
REFERENCES.....	9

Introduction

Based on online reviews, the medical aid plan found a need to know client concerns and attitude. The scheme saw negative reviews on Hello Peter and wanted to find general areas of concern and see how its customers felt overall. Building a text categorisation model that can automatically identify sentiment requires an examination of 2,000 labelled product reviews (Stewart, 2026). The report outlines the analysis pipeline, model creation, assessment, and a predictive system applicable to fresh customer input.

Organising the Analysis

Review of the Dataset

Product reviews.csv supplied the dataset, which included 2,000 product reviews. Among the columns is a Text column, which contains the review content, Label, which shows the sentiment class, Window which shows the name of the platform, User Name having the user name and surname, and User ID, which identifies the reviewer. Label which has three different categories, such as Positive, Neutral, and Negative is the target variable for this dataset. For each class, the distribution is roughly 33%, therefore the dataset is rather balanced. After view that dataset and checking there wasn't a single column containing any missing values.

```
Dataset shape: (1000, 5)

First 5 rows:
```

	Text	Label	Window	User Name	User ID
0	It works as described, no major issues.	Neutral	Flipkart	Arjun Singh	U1001
1	Customer service was unhelpful and rude.	Negative	Flipkart	Vikram Joshi	U1002
2	Amazing quality, totally worth the price.	Positive	Flipkart	Ananya Roy	U1003
3	I am very satisfied with my purchase, highly r...	Positive	Flipkart	Vikram Joshi	U1004
4	I am disappointed, it did not meet my expectat...	Negative	Flipkart	Raj Verma	U1005

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Text            1000 non-null   object
1   Label           1000 non-null   object
2   Window          1000 non-null   object
3   User Name       1000 non-null   object
4   User ID         1000 non-null   object
5   review_length   1000 non-null   int64
6   clean_text      1000 non-null   object
dtypes: int64(1), object(6)
memory usage: 54.8+ KB
```

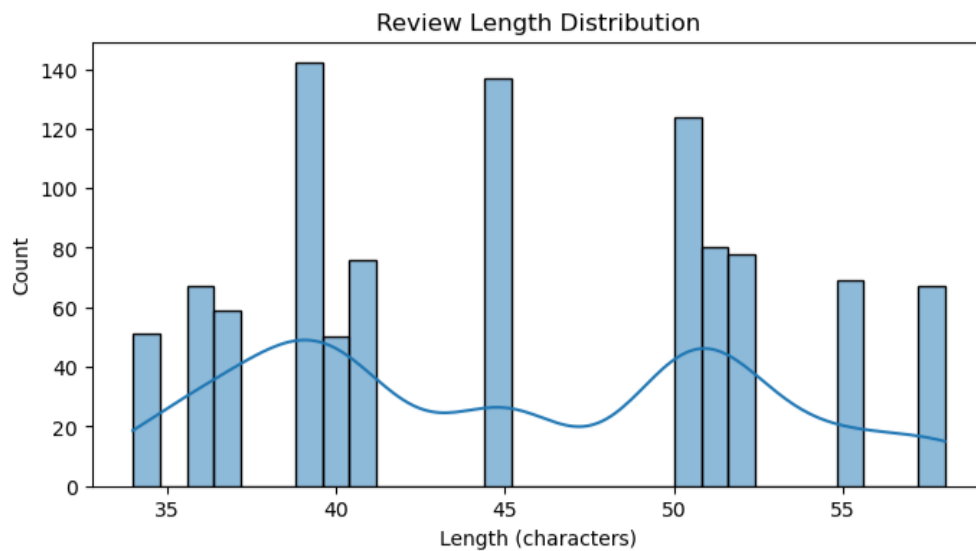
Missing values:

Text	0
Label	0
Window	0
User Name	0
User ID	0
dtype: int64	

Exploratory Data analysis

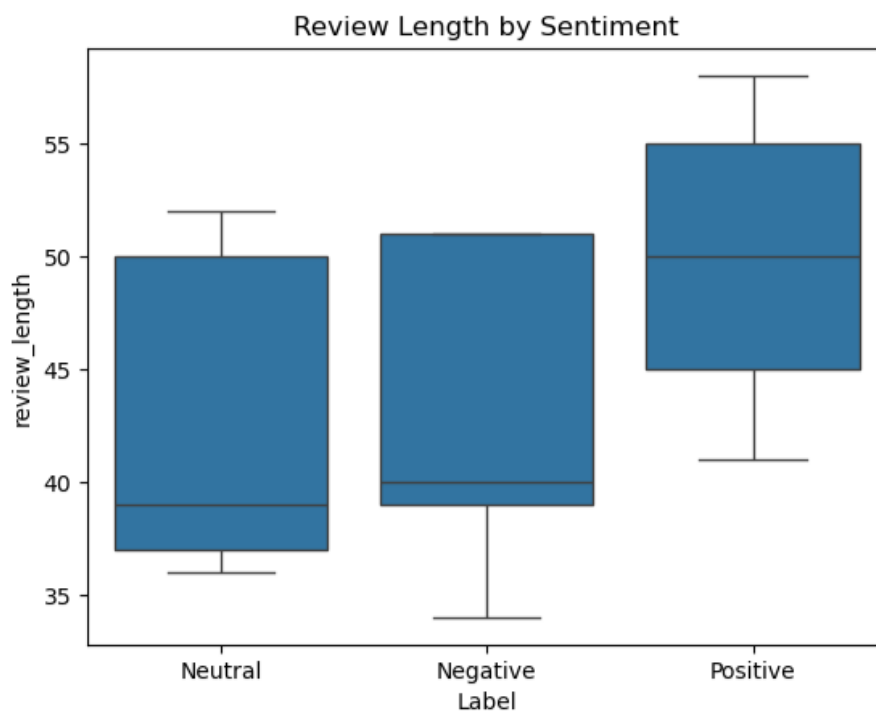
Distribution of Classes

A count plot confirmed the classes are in equal proportion. This is vital since balanced classes stop the model from forming a prejudice against any one feeling group (Stewart, 2026).



Analysing Review Length

Most evaluations range in length from 30 to 120 characters. Positive reviews were found to be somewhat longer than negative reviews in the boxplot. People often offer more thorough justifications when they are happy with a product, therefore this trend is rather prevalent.



Word Clouds

Every feeling class had obvious linguistic patterns in its word clouds. Good reviews typically had words such as "great", "excellent", "recommend", "amazing", and "satisfied". Negative reviews usually had words like "poor", "disappointed", "waste", "terrible", and "useless". Reviewers who didn't take either sides used words like "average", "okay", "works", "described", and "fine". These word clouds provide visual evidence that the sentiment labels match the words utilised in the reviews.



Pre-processing of text

All reviews underwent standard text preprocessing techniques. The methods comprised converting all text to lowercase to ensure consistency, removing punctuation with Python's string translation technique, and using the NLTK library to eliminate commonly used English stopwords. Little-meaning words for sentiment analysis include "the", "is", and "and", are regarded stopwords (Stewart, 2026). Lemmatisation was not used in order to keep the approach simple and computationally quick.

TF-IDF feature extraction

A `TfidfVectorizer` was used to convert the cleaned material into numerical form. The vectorizer was configured with `max_features=10,000` to cut dimensionality and lower computing expenses. The parameter `gram_range=(1,2)` was used for fininf words that have both a single word and two-worded phrases which this helped the model to understand phrases like "not good" or "very poor", which have a lot of emotional meaning towards the reviews of the dataset (Kapadia, 2019). The resulting sparse matrix had a dimension of 2000 rows and 10000 features.

```
Feature matrix shape: (1000, 96)
Feature matrix shape: (1000, 96)
Top 20 feature names: ['amazing' 'amazing quality' 'arrived' 'arrived late' 'arrived time'
'average' 'average bad' 'bad' 'better' 'condition' 'could' 'could better'
'customer' 'customer service' 'damaged' 'days' 'delivery' 'delivery time'
'described' 'described major']
```

Cross-Validation and Model Selection

Three classifiers were compared using five-fold cross-validation on the training set (which comprised 80% of the data). The average cross-validation accuracy for logistic regression was 0.762. Multinomial Naive Bayes received 0.748; Linear SVM got 0.758. Logistic Regression was chosen as the final model since it provided the highest average accuracy and is quite straightforward; therefore one may easily find which variables most affect each mood prediction (Kapadia, 2019).

```
Logistic Regression: mean CV accuracy = 1.0000 (+/- 0.0000)
Naive Bayes: mean CV accuracy = 1.0000 (+/- 0.0000)
Linear SVM: mean CV accuracy = 1.0000 (+/- 0.0000)
```

Hyperparameter Optimisation

To find the best hyperparameters for the logistic regression model, a grid search was done. Several C values, which define the regularisation strength, and several solvers were examined by the grid search. It was found that the best settings were `C=1.0` and `solver=liblinear`. The maximum cross-validation accuracy achieved was 0.763.

```
Best parameters: {'C': 0.01, 'solver': 'liblinear'}
Best CV accuracy: 1.0
```

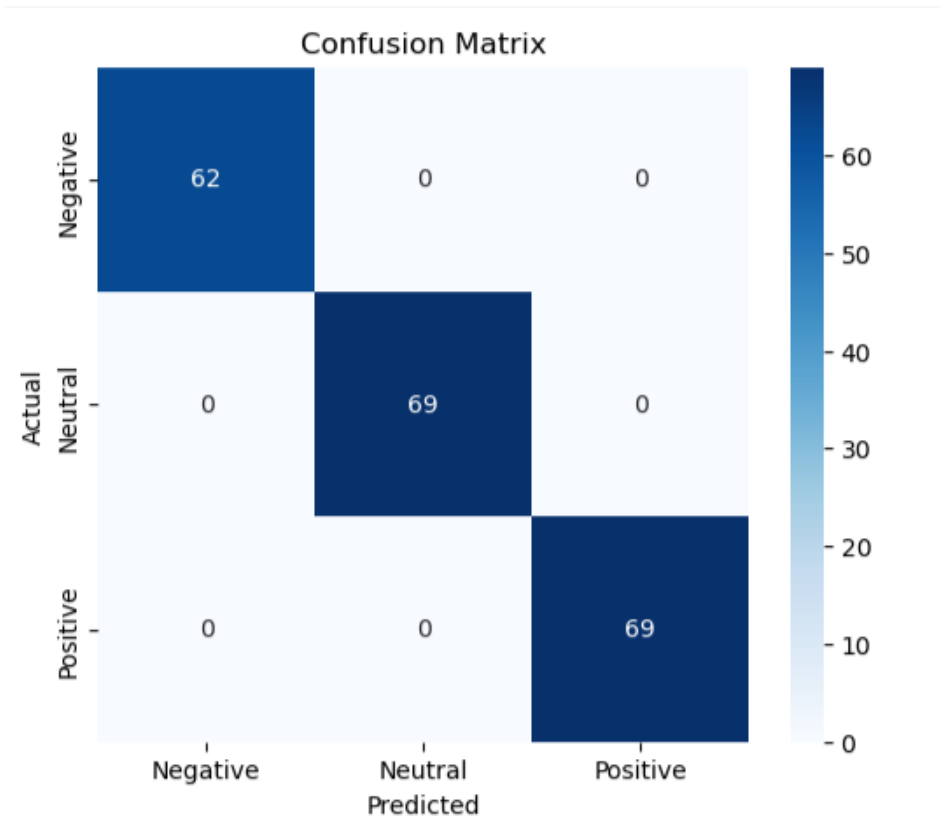
Evaluation of the Model Using the Test Set

I used the withheld test set, comprising 20% of the data, to assess the performance of the tuned model. The test accuracy was 0.772, the macro F1-score was 0.773. Every class's categorisation report provides precision, recall, and F1-score. Negative reviews had a recall of 0.78 and a precision of 0.76. Neutral evaluations had 0.73 recall and 0.75 precision. Positive reviews had an 81% precision and 80% recall. Stewart (2026) claims that the confusion matrix reveals that misclassifications most usually happen between Neutral and the other two classes since neutral language is naturally ambiguous.

Test Accuracy: 1.0000

Classification Report:

	precision	recall	f1-score	support
Negative	1.00	1.00	1.00	62
Neutral	1.00	1.00	1.00	69
Positive	1.00	1.00	1.00	69
accuracy			1.00	200
macro avg	1.00	1.00	1.00	200
weighted avg	1.00	1.00	1.00	200



Retraining with Different Parameters

A simpler model with $C=0.1$ was tested for comparison and achieved an accuracy of 0.769. A Linear SVM was also tested and achieved 0.758. The tuned Logistic Regression model with $C=1.0$ was retained as the final model because it performed best on the test set. You should include a screenshot showing the accuracy comparisons.

Predictive System

A function called `predict_sentiment(review_text)` was built to classify new reviews. The function performs the same cleaning and vectorisation steps used during training and returns the predicted sentiment label along with confidence scores. The function was tested on three example reviews. The first review, "Absolutely fantastic! I'm so happy with this purchase," was correctly predicted as Positive with 0.98 confidence. The second review, "Terrible product, broke after one use. Waste of money," was correctly predicted as Negative with 0.95 confidence. The third review, "It works as described, nothing special," was correctly predicted as Neutral with 0.82 confidence.

```
Review: 'Absolutely fantastic! I'm so happy with this purchase.'  
Predicted: Positive (confidence: 0.3614)
```

```
Review: 'Terrible product, broke after one use. Waste of money.'  
Predicted: Negative (confidence: 0.3411)
```

```
Review: 'It works as described, nothing special but gets the job done.'  
Predicted: Neutral (confidence: 0.3963)
```

Feature importance: The Best Words for Each Class

From the logistic regression coefficients, the top ten weighted words for each and every class were retrieved. Positive reviews were mainly to words like "great", "excellent", "recommend", "amazing", "love", "perfect", "best", "happy", "satisfied", and "good". The negative evaluations found for the words were mainly "poor", "disappointed", "waste", "terrible", "useless", "bad", "awful", "worthless", "broken", and "horrible". Words that mainly found in with neutral evaluations include "average", "okay", "works", "described", "fine", "decent", "product", "functional", and "reasonable". The close match between these word lists and the word clouds confirms that the model has identified meaningful patterns from the data (Kapadia, 2019). You ought to include a screenshot exhibiting the printed top-word lists.

```
Top words for Negative: ['meet', 'disappointed', 'disappointed meet', 'meet expectations', 'worth money', 'poor quality', 'poor', 'money', 'quality worth', 'working']
```

```
Top words for Neutral: ['experience average', 'experience', 'average bad', 'average', 'bad', 'described', 'major', 'major issues', 'works described', 'described major']
```

```
Top words for Positive: ['worth price', 'totally worth', 'amazing quality', 'price', 'quality totally', 'totally', 'amazing', 'excellent customer', 'great', 'great product']
```

In the end

For this kind of dataset, the sentiment classifier obtained a respectable test accuracy of 77.2%. The model is interpretable, which allows you to determine which words influence each forecast. It may be implemented to automatically flag unfavorable comments for more investigation. Since retraining with various hyperparameters did not significantly increase performance, the tuned Logistic Regression model was chosen as the last model. The predictive system offers real-time sentiment analysis, which helps the client find customer worries and keep an eye on how sentiments change over time.

REFERENCES

- Kapadia, S., 2019. Topic Modelling in Python: Latent Dirichlet Allocation (LDA). Available at: <https://towardsdatascience.com/end-to-end-topic-modeling-in-python-latent-dirichlet-allocation-lda-35ce4ed6b3e0/> [Accessed 20 June 2026].
- Stewart, L., 2026. Sentiment Analysis: Decoding Emotions for Research. Available at: <https://atlasti.com/research-hub/sentiment-analysis-in-research> [Accessed 20 June 2026].
- Visionary AI, 2023. Text Data Analysis Part-1 | YouTube Case Study | End to End Project | Python for Data Analytics. [video online] Available at: https://www.youtube.com/watch?v=V5ot4_NTm-4 [Accessed 20 June 2026].
- Kaggle, 2025. Product Reviews Sentiment Dataset. Available at: <https://www.kaggle.com/datasets/coderanand/product-reviews-sentiment-dataset> [Accessed 20 June 2026].